BRAINSTORM GROUP (BSG)

MAT111
Elementary Mathematics I
E-Test Questions (2022-23)

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1. Given $P = \{^{Prime\ number\ between}_{10\ and\ 15}\}$ and $Q = \{factor\ of\ 15\}$, find $P \cap Q$
   A. $\{3, 5, 11, 13, 15\}$ B. $\emptyset$
   C. $\{11, 13\}$ D. $\{15\}$

2. A die is thrown once, what is the value of all possible subsets on the die?
   A. 60 B. 62
   C. 64 D. 68

3. List the element of the set $Y = \{x: x^2 - 9 = 0; x > 0\}$
   A. $\{-3, 3\}$ B. $\{\ \}$
   C. $\{3\}$ D. $\{-3\}$

4. Given the set $X = \{1,\ 2,\ 3\}$ and set $Y = \{3,\ 1,\ 2\}$, which of the following statement is true for $X\ and\ Y$?
   A. $X \neq Y$ B. $X \leq Y$
   C. $X \geq Y$ D. $X = Y$
   E. $X \equiv Y$

5. Determine the zero of the function $2x - 3$
   A. $-\frac{3}{2}$ B. -3
   C. $\frac{3}{2}$ D. $-\frac{2}{3}$

6. List $n(even\ numbers)$
   A. 100 B. 200
   C. infinite D. 400

7. Find imaginary part of $\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2$
   A. $-\frac{1}{2}\sqrt{3}i$ B. $\frac{1}{2}\sqrt{3}i$
   C. $-\frac{1}{4}\sqrt{3}i$ D. $\frac{1}{4}\sqrt{3}i$

8. Given $Z = 4 - 2\sqrt{2}i,\ find\ Z\bar{Z}$
   A. 20 B. 22
   C. 24 D. 21

9. Given $Z_1 = (3 + 2i)\ and\ Z_2 = (-1 + i), evaluate\ Z_1 + Z_2$
   A. $4 - 3i$ B. $2 - 3i$
   C. $4 + 2i$ D. $2 + 2i$

10. In an examination, 18 students passed MAT111, 17 students passed PHY113, 11 students passed both subject. Find the number of students that passed MAT111 only.
    A. 5 B. 6
    C. 7 D. 8

11. If $Z_1 = 1 - 3i$ and $Z_2 = -3 - 4i, find\ \frac{Z_1}{Z_2}$
    A. $\frac{9+11i}{25}$ B. $\frac{-9+11i}{25}$
    C. $\frac{9-11i}{25}$ D. $\frac{9+11i}{5}$

12. If $(1 + 2i)(-2 - 3i) = x + iy$, find x and y respectively
    A. $-7\ and\ 4$ B. $-7\ and\ -4$
    C. $-4\ and\ -7$ D. $4\ and\ -7$

13. Simplify $\sqrt{27} + \sqrt{12} - \sqrt{48}$
    A. $2\sqrt{3}$ B. $-\sqrt{3}$
    C. $-3\sqrt{3}$ D. $\sqrt{3}$

14. If $X = \{2, 4, 6\}$, determine power set
    A. 8 B. 6
    C. 4 D. 3

15. Find the cardinality of set $P = \{x: 3 < x \leq 20\}$
    A. 16 B. 17
    C. 18 D. 19

16. Find the cardinality of $n\ \epsilon\ \mathbb{N} : 3n > 12$
    A. 4 B. 3
    C. infinity D. $\{0, 1, 2, 3, 4\}$

17. What is the value of $n(\{3, 5\})$
    A. 1 B. 2 C. 3 D. 4

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18. List the element of the set $X = \{n^2 + 1;\ n = 0, 1, 2, 3\}$

A. $\{0, 2, 4, 9\}$

B. $\{1, 2, 5, 10\}$

C. $\{0, 1, 2, 3\}$

D. $\{0, 2, 5, 10\}$

19. Find $n(\{1, \{2, 3\}, \{4, 5, 6\}\})$

A. 1 B. 2

C. 3 D. 6

20. Given the universal set $\mathcal{U} = \{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$ where $X = \{2,\ 4,\ 6\}$ and $Y = \{1,\ 2,\ 6\}$, find $(X \cap Y)^c$

A. $\{2, 4, 5, 6\}$

B. $\{1,\ 3,\ 4,\ 5\}$

C. $\{3,\ 5,\ 6,\ 7\}$

D. $\{1,\ 4,\ 5\}$

E. $\{3,\ 4,\ 5\}$

21. Given that $S = \{All\ states\ in\ Nigeria\}$, what is the cardinality of S?

A. 34 B. 36

C. 37 D. 38

22. If $\sqrt{x} - \sqrt{x-2} = 1$, find the positive values of x which satisfies the equation.

A. $\frac{3}{4}$ B. $\frac{5}{4}$

C. $\frac{7}{4}$ D. $\frac{9}{4}$

23. Given $P = \{even\ numbers\}$ and $Q = \{x : 7 < x \leq 16$. List the elements of $P \cap Q$

A. $\emptyset$

B. $\{8, 10, 12, 14\}$

C. $\{8, 10, 12\}$

D. $\{8, 10, 12, 14, 16\}$

24. If $Z_1 = 1 - 3i$ and $Z_2 = -2 + 5i$, determine $\frac{Z_1 Z_2}{Z_1 + Z_2}$ in the form $a + ib$

A. $-\frac{9}{5} + \frac{37}{5}i$ B. $\frac{9}{5} - \frac{37}{5}i$

C. $-\frac{12}{25} - \frac{13}{25}i$ D. $\frac{9}{25} - \frac{37}{25}i$

25. Given the universal set $U = \{1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8\}$ where $A = \{1,\ 3,\ 5,\ 7\}, B = \{1,\ 2,\ 5,\ 7\}$ and $C = \{3,\ 6,\ 7,\ 8\}$. List the members of $A^c \cap B \cap C^c$

A. $\{1\}$ B. $\{2\}$ C. $\{3\}$

D. $\{4\}$ E. $\{5\}$

26. If $F = \{x: (2 < x \leq 5) \cup (9 < x < 12)\}$, list the elements of set F

A. $\{2, 3, 4, 5, 9, 10, 11, 12\}$

B. $\{3, 4, 5, 9, 10, 11\}$

C. $\{3, 4, 5, 10, 11\}$

D. $\{3, 4, 5, 10, 11, 12\}$

27. Find the real part of $\frac{2+3i}{3+2i}$

A. $\frac{12}{13}$ B. $-\frac{5}{13}$

C. $\frac{5}{13}$ D. $-\frac{12}{13}$

28. If $Z = cos\theta + isin\theta$, then find the value of $\left(z + \frac{1}{z}\right)^2$ is

A. $2cos\theta$ B. $2isin\theta$

C. $4cos^2\theta$ D. $-4sin^2\theta$

29. Express $cos^2\theta$ is equal to

A. $2(cos2\theta + 1)$

B. $2^{-1}(cos2\theta + 1)$

C. $2^{-1}(cos2\theta - 1)$

D. $2(cos2\theta - 1)$

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30. Given $f(x) = x + 1$ and $g(x) = (x + 1)^2$, find gof

A. $x^2 + 4x + 2$

B. $x^2 + 4x + 4$

C. $x^2 + 2x + 4$

D. $x^2 + 2x + 2$

31. Find the value of a if $a + ib = \frac{1-i}{2+i}$

A. $-\frac{1}{5}$ B. $\frac{3}{5}$

C. $-\frac{3}{5}$ D. $\frac{1}{5}$

32. Find the range of $y = 7x + 2$

A. $-\infty, 0$ B. $0, \infty$

C. $-\infty, \infty$ D. $0, 1$

33. Solve the equation $\sqrt{3x + 4} - \sqrt{x + 5} = 1$

A. -4 B. 4

C. 1 D. -1

34. Which of the following set description is not correct?

A. $B = set\ of\ Geometry\ book$

B. $A\ \{x; even\}$

C. $H\ \{1, 2, 3: 4\}$

D. $S = \{a, b, c\}$

35. What is the value of $n(\{\})$

A. 0 B. 1

C. 2 D. 3

36. What is the value of $n(\{a, a\})$

A. 0 B. 1

C. 2 D. 3

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ANSWERS TO MAT111							
1.	B	11.	A	21.	B	31.	D
2.	C	12.	D	22.	D	32.	C
3.	C	13.	D	23.	D	33.	B
4.	D	14.	A	24.	B	34.	C
5.	C	15.	B	25.	B	35.	A
6.	C	16.	C	26.	C	36.	B
7.	A	17.	B	27.	A		
8.	C	18.	B	28.	C		
9.	D	19.	C	29.	B		
10.	C	20.	B	30.	B		

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